

PAPER_677: UQFF Predictions for the LISA Space-Based Gravitational Wave Observatory

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Framework: Unified Quantum Field Framework (UQFF) v5.30

Session: 173

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Class: UQFFPredictionsForLISA | CP4 #261

Source: grok_share_fc21e30c24b4.txt

Domain: Space-Based GW / LISA

Abstract

The Laser Interferometer Space Antenna (LISA) will open the millihertz GW window, targeting SMBH mergers, EMRIs, and galactic binaries. The UQFF introduces a unique LISA-band suppression via the Aether arm-length modulation: $S_{\text{UA,LISA}} = 1 - \rho_{\text{UA}} L_{\text{LISA}} / (k_B T_{\text{eff}})$ and predicts an enhanced EMRI capture rate $R_{\text{EMRI,UQFF}} = R_{\text{GR}} (1 + f_{\text{TRZ}} \rho_{\text{UA}} / \rho_{\text{SCm}})$.

Primary UQFF Equation

$$h_{\text{UQFF,LISA}}(f) = h_{\text{GR}}(f) \cdot (1 - f_{\text{TRZ}}) \cdot S_{\text{UA,LISA}} \cdot S_{\text{SCm}}$$

Parameters

Parameter	Value	Description
L_{LISA}	2.5 Gm	LISA arm length
f	10^{-4} -1 Hz	LISA band
M_c	$10^8 M_{\odot}$	SMBH chirp mass

UQFF Constants

Constant	Value	Description
ρ_{UA}	$7.09 \times 10^{-36} \text{ kg/m}^3$	Universal Aether density
ρ_{SCm}	$7.09 \times 10^{-37} \text{ kg/m}^3$	Schwarzschild condensate
f_{TRZ}	0.1	Time-reversal zone factor
κ	$5 \times 10^{-4} \text{ day}^{-1}$	UQFF calibration constant
[SSq]	0.57	Superstring quenching factor
μ_J	$3.38 \times 10^{23} \text{ J}\cdot\text{m}$	Magnetic string coupling
γ	$5 \times 10^{-5} / 86400 \text{ s}^{-1}$	Aether oscillation decay

C++ Implementation

```
#include "UQFFPredictionsForLISA.h"
```

```
UQFFPredictionsForLISA obj;  
auto result = obj.compute_primary(...); // primary equation  
obj.simulate_over_M(M_start, M_end, dM, "output.csv");
```

Python CP4 Calculator

```
from CondensedPhysics2 import UQFFPredictionsForLISACalculator
```

```
calc = UQFFPredictionsForLISACalculator()  
result = calc.compute() # returns all UQFF quantities  
print(result)
```

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	✓ Consistent
Cosmological constant Λ	$1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)	$1.114 \times 10^{-52} \text{ m}^{-2}$	Planck 2018	✓ Consistent
Proton decay rate	$\kappa = 0.0005/\text{day} \rightarrow \Gamma_{\text{p}}$ suppression	$< 4.17 \times 10^{-35}/\text{yr}$	Super-K 2024	✓ Consistent
UQFF buoyancy signature	F_U_Bi_i unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

References

Amaro-Seoane et al. 2017 (LISA); UQFF Session 173

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